THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 70	Area $= \frac{1}{2} \times a \times b$.
2	**(b)** 25	Intercepts are c and $-c/m$; area $= \frac{1}{2}\lvert c\rvert \cdot \lvert c/m\rvert$.
3	**(a)** $\frac{7}{9}$	Each is $\frac{d}{9}$.
4	**(a)** 35	$x^3 + y^3 = (x+y)^3 - 3xy(x+y)$.
5	**(c)** 5	The circumcentre is the midpoint of the hypotenuse; its distance from the right-angle vertex is half the hypotenuse.
6	**(d)** 4	Equate the two expressions.
7	**(d)** $\frac{4}{9}$	Add the numerators over 99.
8	**(b)** 98	$x^3 - y^3 = (x-y)^3 + 3xy(x-y)$.
9	**(d)** 41	Radius $= \frac{1}{2}\times$ distance between the endpoints.
10	**(c)** 25	Evaluate the inner value first.
11	**(c)** 3	Cancel every prime factor other than 2 and 5.
12	**(d)** 18	$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$.
13	**(a)** Both A and R are true and R is the correct explanation of A.	$x = -3 < 0$ and $y = 4 > 0$, which is exactly the sign pattern of Quadrant II.
14	**(a)** Both A and R are true and R is the correct explanation of A.	$-\dfrac{-12}{3} = 4$; R is the general rule that gives it.
15	**(a)** Both A and R are true and R is the correct explanation of A.	$0.375 = \dfrac{375}{1000} = \dfrac{3}{8}$; R is the general reason.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Circumcentre is the midpoint of $AB = (3,4)$ (1); Radius $= OM = \sqrt{9+16} = 5$ (1).	1+1
17	$200 + 30x = 440 + 10x$ (1); $x = 12$ units (1).	1+1
18	$0.\overline{17} = \frac{17}{99}$, $0.\overline{71} = \frac{71}{99}$ (1); Sum $= \frac{8}{9}$ (1).	1+1
19	$x^3 + y^3 = (x+y)^3 - 3xy(x+y)$ (1); $= 512 - 288 = 224$ (1).	1+1
20	Centre $= (7,9)$ (1); $AB = 2\sqrt{36+64} = 20$, radius $= 10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Let the point be $(x,0)$: $(x-2)^2 + 400 = 625$ (1); $(x-2)^2 = 225$ (1); $x = 17$ or $x = -13$ (1).	1+1+1
22	$5x + 2 = 2x + 11$ (1); $x = 3$ (1); $y = 17$ (1).	1+1+1
23	$x = 0.4\overline{7}$: $100x - 10x = 43$ (1); $x = \frac{43}{90}$ (1); Long division of p by q gives $0.4\overline{7}$ (1).	1+1+1
24	$\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$ (1); $x^3 + \frac{1}{x^3} = 7^3 - 3(7)$ (1); $= 322$ (1).	1+1+1
	OR: $C = B + D - A = (-7,24)$ (1); $AB = 15$, $AD = 20$ (1); $AC = \sqrt{225+400} = 25$ (1).	1+1+1
25	$p(3) - p(1) = 2a = 8$, so $a = 4$; $b = 8$, $p(x) = 4x + 8$ (1); Zero $x = -2$ (1); $p(6) = 32$ (1).	1+1+1
	OR: $\frac{1}{x} = 2 - \sqrt{3}$ since $2^2 - 3 = 1$, so $x + \frac{1}{x} = 4$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 16 - 2 = 14$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $x < 0$, $y > 0$, so Quadrant II (1). (ii) $(-3,-4)$ (1). (iii) Same abscissa, so $AB = \lvert 4 - (-2)\rvert = 6$ km (1). (iv) Midpoint $= \left(\dfrac{-3+5}{2}, \dfrac{4-2}{2}\right) = (1,1)$ (1); $AC = \sqrt{8^2 + 6^2} = 10$ km (1).	1+1+1+2

27	Divide 3 by 7 until a remainder repeats (1); $\frac{3}{7} = 0.\overline{428571}$ (1); Block 428571 (1); Block length $= 6$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: $x^2 + y^2 = 9 + 56 = 65$ (1); $x^3 - y^3 = (x-y)^3 + 3xy(x-y) = 27 + 252$ (1); $x^3 - y^3 = 279$ (1); $(x+y)^2 = (x-y)^2 + 4xy = 9 + 112$ (1); $(x+y)^2 = 121$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 1030301	$101 = 100 + 1$; use $(a \pm b)^3$.
2	**(c)** 4	Solve $(x - x_0)^2 + b^2 = c^2$ and take the larger root.
3	**(a)** 7	$kz - c = 0$.
4	**(c)** $\frac{14}{11}$	Write as $1 + 0.\overline{ab}$.
5	**(a)** 21	$2xy = (x + y)^2 - (x^2 + y^2)$.
6	**(b)** $(3, 4)$	$D = A + C - B$.
7	**(b)** $b = 0$	Reason about linear polynomials.
8	**(b)** 1	Reason about decimal expansions.
9	**(b)** 28	$2xy = (x^2 + y^2) - (x - y)^2$.
10	**(d)** 3	Solve $(x - a_0)^2 + b^2 = c^2$.
11	**(d)** 64	Find each term.
12	**(a)** 5	Find the order of 10 modulo q.
13	**(a)** Both A and R are true and R is the correct explanation of A.	$103^2 = (100 + 3)^2 = 10000 + 600 + 9 = 10609$, using R.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	A point with abscissa 0 is on the y-axis, so A is true; R is also true but is a statement about the x-axis, so it does not explain A.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Putting $y = 0$ gives $x = 3$, so A is true; putting $x = 0$ gives $y = -6$, so R is true. R gives the other intercept, so it does not explain A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$p(p(x)) = 3(3x + 4) + (4) = 9x + 16$ (1); Value $= 34$ (1).	1+1
17	$420 = 2^2 \times 3 \times 5 \times 7$ (1); Cancel the factors other than 2 and 5: $n = 21$ (1).	1+1
18	$103 = 100 + 3$ (1); $103^3 = 1092727$ (1).	1+1
19	$d^2 = (9)^2 + (-12)^2 = 225$ (1); $d = 15$ (1).	1+1
20	Intercepts: $y = 14$ and $x = -7$ (1); Area $= \frac{1}{2} \times 14 \times 7 = 49$ sq. units (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Need two numbers with sum -5 and product 4; They are -1 and -4 (1); $= (x - 1)(x - 4)$ (1); At $x = 6$: $(5)(2) = 10$ (1).	1+1+1
22	$D = (21, 28)$ (1); $AD = \sqrt{441 + 784}$ (1); $AD = 35$ units (1).	1+1+1
23	$14n + 3 < 143$ gives $n < 10$ (1); Terms: $n = 1, 2, \ldots, 9$, number $= 9$ (1); $a_1 + a_2 = 17 + 31 = 48$ (1).	1+1+1
24	Suppose $3 + \sqrt{2} = r$ is rational; Then $\sqrt{2} = r - 3$ (1); $r - 3$ is rational, so $\sqrt{2}$ would be rational (1); This contradicts the given fact (1).	1+1+1
	OR: Use $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ (1); $a = 2x$, $b = 3$ (1); $= (2x - 3)(4x^2 + 6x + 9)$ (1).	1+1+1
25	Let the point be $(x, 0)$: $(x + 1)^2 + 576 = 676$ (1); $(x + 1)^2 = 100$ (1); $x = 9$ or $x = -11$ (1).	1+1+1
	OR: $7x - 3 = 3x + 9$ (1); $x = 3$ (1); $y = 18$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $C(x) = 12x + 50$ (1). (ii) $C(10) = 120 + 50 =$ Rs 170 (1). (iii) $12x + 50 = 230 \Rightarrow x = 15$ km (1). (iv) Slope $= 12$: the rate of Rs 12 per km (1); y-intercept $= 50$: the fixed charge of Rs 50 (1).	1+1+1+2

27	First $= 2(7+5) = 24$ (1); Second $= 7-5 = 2$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{35}$ cancel (2).	1+1+1+2
	OR: $2xy = (x+y)^2 - (x^2+y^2) = 100 - 58 = 42$ (1); $xy = 21$ (1); $(x-y)^2 = x^2+y^2-2xy = 58-42 = 16$ (1); $x-y = 4$ (1); $x^2-y^2 = (x+y)(x-y) = 40$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 2	Multiply the fraction by 11 and reduce.
2	**(c)** $(2x+3)(4x^2-6x+9)$	Use the factorisation identity.
3	**(c)** 72	Area $=\frac{1}{2}\times a\times b$.
4	**(a)** 18	Intercepts are c and $-c/m$; area $=\frac{1}{2}\lvert c\rvert\cdot\lvert c/m\rvert$.
5	**(a)** $\frac{4}{3}$	Each is $\frac{d}{9}$.
6	**(a)** 133	$x^3+y^3=(x+y)^3-3xy(x+y)$.
7	**(d)** 13	The circumcentre is the midpoint of the hypotenuse; its distance from the right-angle vertex is half the hypotenuse.
8	**(d)** 5	Equate the two expressions.
9	**(d)** $\frac{8}{9}$	Add the numerators over 99.
10	**(b)** 117	$x^3-y^3=(x-y)^3+3xy(x-y)$.
11	**(d)** 37	Radius $=\frac{1}{2}\times$ distance between the endpoints.
12	**(c)** 14	Evaluate the inner value first.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but R is a property of irrational numbers and does not prove that $\sqrt{2}$ is irrational.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	A follows from $a^2-b^2=(a-b)(a+b)$; R is true but is a different identity and does not explain A.
15	**(c)** A is true but R is false.	$x>0,\ y<0$ gives Quadrant IV, so A is true. In Quadrant IV the abscissa is positive, so R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\frac{1}{x}=\frac{6-\sqrt{35}}{6^2-35}=6-\sqrt{35}$ (1); $x+\frac{1}{x}=12$ (1).	1+1
17	Use $a^3+b^3=(a+b)(a^2-ab+b^2)$; $a=2x,\ b=1$ (1); $=(2x+1)(4x^2-2x+1)$ (1).	1+1
18	Circumcentre is the midpoint of $AB=(5,12)$ (1); Radius $=OM=\sqrt{25+144}=13$ (1).	1+1
19	$150+25x=390+5x$ (1); $x=12$ units (1).	1+1
20	$0.\overline{29}=\frac{29}{99}$, $0.\overline{92}=\frac{92}{99}$ (1); Sum $=\frac{11}{9}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$x=5.3\overline{2}$: $100x-10x=479$ (1); $x=\frac{479}{90}$ (1); Long division of p by q gives $5.3\overline{2}$ (1).	1+1+1
22	$\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=8^3-3(8)$ (1); $=488$ (1).	1+1+1
23	$C=B+D-A=(1,27)$ (1); $AB=20,\ AD=15$ (1); $AC=\sqrt{400+225}=25$ (1).	1+1+1
24	$p(3)-p(1)=2a=10$, so $a=5$; $b=-15$, $p(x)=5x-15$ (1); Zero $x=3$ (1); $p(4)=5$ (1).	1+1+1
	OR: $\frac{1}{x}=3-\sqrt{8}$ since $3^2-8=1$, so $x+\frac{1}{x}=6$ (1); $x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2$ (1); $=36-2=34$ (1).	1+1+1
25	Need two numbers with sum -9 and product 14; They are -2 and -7 (1); $=(x-2)(x-7)$ (1); At $x=10$: $(8)(3)=24$ (1).	1+1+1
	OR: $D=(24,32)$ (1); $AD=\sqrt{576+1024}$ (1); $AD=40$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	(i) $\frac{7}{16} = 0.4375$ (1). (ii) $\frac{11}{25} = 0.44$ (1). (iii) $12 = 2^2 \times 3$ (1). (iv) $\frac{7}{16}$ and $\frac{11}{25}$ terminate (1) because $16 = 2^4$ and $25 = 5^2$ have only the prime factors 2 and 5; $\frac{5}{12} = 0.41\overline{6}$ does not terminate because 12 has the factor 3 (1).	1+1+1+2
27	Divide 5 by 13 until a remainder repeats (1); $\frac{5}{13} = 0.\overline{384615}$ (1); Block 384615 (1); Block length $= 6$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: $x^2 + y^2 = 4 + 48 = 52$ (1); $x^3 - y^3 = (x-y)^3 + 3xy(x-y) = 8 + 144$ (1); $x^3 - y^3 = 152$ (1); $(x+y)^2 = (x-y)^2 + 4xy = 4 + 96$ (1); $(x+y)^2 = 100$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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